

Unit 7: Game Theory 7 Hrs.

Introduction, Characteristics of Game Theory, Two Person, Zero sum games, pure strategy. Dominance theory, Mixed strategies (2x2, mx2), Algebraic and graphical methods

#Past Questions

- **2021 Group C-(11 b).** Explain the theory of two-person zero-sum games with an example.
- **2023 11 a.** Two types of food manufacturers, A and B, are competing for an increased market share. The payoff matrix describes the increase in market share for A and the decrease in market share for B. Determine the optimal strategies for both manufacturers and the value of the game.

A	B			
	Give Prizes	Decrease Price	Maintain Present Strategy	Increase Advertising
Give Prizes	25	20	14	30
Decrease Price	27	16	12	14
Maintain Present Strategy	35	8	15	19
Increase Advertising	-2	8	13	0

- **2024 Q4.** Reduce the following two-person zero sum game to 2x2 order by dominance rule and obtain the optimal strategies for each player and the value of the game.

Player A	Player B			
	B1	B2	B3	B4
A1	3	2	4	0
A2	3	4	2	4
A3	4	2	4	0
A4	0	4	0	8

- **2024 Group C –(9).** What is game theory? State its assumptions and business-decision importance.

Introduction

Game is defined as an activity among two or more persons as per a set of rules at the end of which each person gets some benefit or bears loss.

- The set of rules and procedure defines the game.
- A competitive situation is called a game.
- The term *game* represents a conflict between two or more parties.

Real-life Examples of Games:

- **Business:** Competing brands deciding on advertising strategies.
- **Politics:** Countries negotiating trade or peace deals.
- **Everyday life:** Deciding whether to cooperate or cheat in group tasks.

Game Theory:

- Game theory studies interactive decision-making where the outcome for each participant or player depends on the actions of all. If you are a player in such a game, when choosing your course of action or strategy, you must take into account the choices of others.
- Game Theory is a branch of operations research and mathematics that deals with decision-making in situations of competition or conflict. It analyzes strategies where the outcome for each participant depends on the choices of all.
- Game Theory is the study of strategic interactions among rational decision-makers (called players), where the outcome (payoff) for each player depends on the strategies chosen by all.

- Game Theory is a branch of mathematics and economics that studies how people or organizations behave in strategic situations—i.e., situations where the outcome for each depends on the choices of others.

It helps answer “**What should I do, knowing that others are also trying to make the best choices for themselves?**”

Game theory uses

Game theory helps businesses to predict their competitors' behaviors. Businesses make decisions based on a set of strategies and their decisions can affect overall revenue. Economists can use this theory to predict how big firms may act or react to situations like price wars and the likely outcomes.

Types of Games in Game Theory:

1. **Cooperative vs. Non-Cooperative Games:**
 - *Cooperative:* Players can form binding agreements (e.g., business coalitions).
 - *Non-cooperative:* No enforceable agreements, only individual actions matter.
2. **Zero-Sum vs. Non-Zero-Sum Games:**
 - *Zero-sum:* One player's gain is another's loss.
Example: Chess, poker.
 - *Non-zero-sum:* All players can benefit or suffer.
Example: Environmental policies.
3. **Simultaneous vs. Sequential Games:**
 - *Simultaneous:* Players act at the same time.
Example: Rock-paper-scissors.
 - *Sequential:* Players take turns.
Example: Chess, or bidding in auctions.
4. **Perfect vs. Imperfect Information:**
 - *Perfect:* All players know past moves.
 - *Imperfect:* Some actions or data are hidden.

Why is Game Theory Important?

- Helps in **decision-making** in complex, competitive situations.
- Widely used in **economics, military strategy, politics, biology, psychology, and computer science.**
- Provides a framework to predict **likely outcomes** and **optimal strategies.**

Famous Examples in Game Theory:

1. **Prisoner's Dilemma:** Two criminals are caught. They can either betray each other or stay silent. The best collective outcome is if both stay silent, but rational self-interest often leads them to betray.
2. **Nash Equilibrium:** A situation where no player can benefit by changing their strategy while others keep theirs unchanged. Named after John Nash (portrayed in *A Beautiful Mind*).

Characteristics/key concepts of Game Theory

- **Players:** Decision-makers or competitors in the game (may be individual or organization, usually two in basic models).
- **Strategies:** Set of actions available to each player (a move chosen by a player).
- **Pure strategy:** If a player selects the same strategy each time.
- **Mixed strategy:** If a player selects the strategy according to some fixed probabilities.
- **Optimal strategy:** A strategy by which a player can achieve the best payoff (that most benefits a player)
- **Payoff:** The outcome/result each player receives based on the chosen strategies.

- **Value of the Game :** It refers to the expected outcome of the game, when players follow their optimal strategy. It is generally denoted by V .
- **Rationality:** Players aim to maximize their own benefit.
- **Saddle Point:** The point where Maximin = Minimax

Minimax and Maximin Principles

- **Maximin principle :** For player A, Select the minimum element from each row and write them in Row Minimum column. Now Select the maximum element from Row Minimum column and enclose it in rectangle []. It is called Row MaxiMin.
- **Minimax principle :** For player B, Select the maximum element from each column and write them in Column Maximum row. Now Select the minimum element from Column Maximum row and enclose it in circle (). It is called Column MiniMax.

Assumptions of Game Theory

1. There are finite number of competitors (players)
2. Each player tries to maximize his gains and minimize his losses
3. Each player has finite number of possible courses of action
4. The choices of action are assumed to be made simultaneously, so that no player knows his competitors move until decided
5. The outcome of game is predetermined and fixed

Two-Person Zero-Sum Games

- A special type of game where:
 - There are two players.
 - One player's gain is exactly equal to the other's loss.
 - Total payoff = 0.

What Does “Two-Person Zero-Sum” Mean?

- **Two-Person:** Only two players are involved in the game.
- **Zero-Sum:** The total gain or loss between both players is always zero.
 ➡ One player's gain is exactly equal to the other player's loss.

Example: Let's say:

- Player A gains +5 from a certain move.
- Then Player B must lose −5.

So, **Payoff(A) + Payoff(B) = 0**

The interests of both players are **completely opposed**.

Real-Life Examples:

1. **Chess** – One player win (+1), the other loses (−1).
2. **Poker (heads-up)** – Money won by one is lost by the other.
3. **Tug of War** – One side pulls the flag to their side (+), other side loses (−).

Assumptions for two-person zero sum game:

- Each player has a **finite number of strategies**.
Some strategies may be **common**, while others may be **unique** to each player.
- **Player A aims to maximize** his gain; **Player B aims to minimize** his loss.
- Both players **make decisions independently**, without communication.
- Decisions are made and **announced simultaneously**, so neither player has an advantage.
- Both players **know all possible payoffs**, for themselves and their opponent.

Types of Two-Person Zero-Sum Game

Basically, there are two types of Two-Person Zero-Sum Game:

1. The game with Saddle Point (Pure Strategy)
2. The game without Saddle Point (Mixed Strategy)

1. The game with Saddle Point (Pure Strategy)

saddle point Steps (Rule)	
Step-1:	1. Select the minimum element from each row and write them in Row Minimum column. 2. Select the maximum element from Row Minimum column and enclose it in []. It is called Row MaxiMin.
Step-2:	1. Select the maximum element from each column and write them in Column Maximum row. 2. Select the minimum element from Column Maximum row and enclose it in (). It is called Column MiniMax.
Step-3:	1. Find out the elements that is same in rectangle [] and circle (). 2. If Column MiniMax = Row MaxiMin then the game has saddle point and it is the value of the game.

Example-1

1. Find Solution of game theory problem using saddle point

Player A \ Player B	B1	B2	B3	B4
A1	20	15	12	35
A2	25	14	8	10
A3	40	2	10	5
A4	-5	4	11	0

Solution:

1. Saddle point testing

		Player B					
		B1	B2	B3	B4		
Player A	A1	20	15	12	35		
	A2	25	14	8	10		
	A3	40	2	10	5		
	A4	-5	4	11	0		

We apply the maximin (minimax) principle to analyze the game.

		Player B				
		<i>B1</i>	<i>B2</i>	<i>B3</i>	<i>B4</i>	Row Minimum
Player A	A1	20	15	[(12)]	35	[12]
	A2	25	14	8	10	8
	A3	40	2	10	5	2
	A4	-5	4	11	0	-5
	Column Maximum	40	15	(12)	35	

Select minimum from the maximum of columns

∴ Column MiniMax = (12)

Select maximum from the minimum of rows

∴ Row MaxiMin = [12]

Here, Column MiniMax = Row MaxiMin = 12

∴ This game has a saddle point and value of the game is 12

The optimal strategies for both players are:

The player A will always adopt strategy A1

The player B will always adopt strategy A3

Example-2

Find Solution of game theory problem using saddle point

Player A \ Player B	B1	B2	B3
A1	-2	14	-2
A2	-5	-6	-4
A3	-6	20	-8

Solution:

1. Saddle point testing

		Player B				
		<i>B1</i>	<i>B2</i>	<i>B3</i>		
Player A	A1	-2	14	-2		
	A2	-5	-6	-4		
	A3	-6	20	-8		

We apply the maximin (minimax) principle to analyze the game.

		Player B			
		<i>B1</i>	<i>B2</i>	<i>B3</i>	Row Minimum
Player A	A1	[(-2)]	14	-2	[-2]
	A2	-5	-6	-4	-6
	A3	-6	20	-8	-8
	Column Maximum	(-2)	20	-2	

Select minimum from the maximum of columns

Column MiniMax = (-2)

Select maximum from the minimum of rows

Row MaxiMin = [-2]

Here, Column MiniMax = Row MaxiMin = -2

∴ This game has a saddle point and value of the game is -2

The optimal strategies for both players are:

The player A will always adopt strategy 1

The player B will always adopt strategy 1

2. The game without Saddle Point (Mixed Strategy)

➤ Used when no saddle point exists.

➤ Game without Saddle Point

If Maximin $\neq$ Minimax

It is also known as mixed strategy problem

➤ **Mixed strategy:** A player chooses more than one strategy with some fix probabilities.

This type of game can be solved by any of the following methods:

- i. Algebraic Method
- ii. Arithmetic Method
- iii. Graphical Method
- iv. Dominance Method

i. Algebraic Method

Consider the zero sum two-person game given below:

		Player B	
Player A		B1	B2
	A1	a	b
	A2	c	d

algebraic method Steps (Rule)

Step-1:	A play's (p_1, p_2) $p_1 = \frac{d - c}{(a + d) - (b + c)} \text{ and } p_2 = 1 - p_1$
Step-2:	B play's (q_1, q_2) $q_1 = \frac{d - b}{(a + d) - (b + c)} \text{ and } q_2 = 1 - q_1$
Step-3:	Value of the game V $V = \frac{a \cdot d - b \cdot c}{(a + d) - (b + c)}$

Example-1: Find Solution of game theory problem using algebraic method

Player A \ Player B	B1	B2
A1	1	7
A2	6	2

Solution:

1. Saddle point testing

		Player B		
		B1	B2	Row Minimum
Player A	A1	1	7	1
	A2	(6)	[2]	[2]
Column Maximum		(6)	7	

Here, Column MiniMax $\neq$ Row MaxiMin

$\therefore$ This game has no saddle point.

Matrix size is 2×2 , so dominance rule is not required.

Solution using algebraic method

Here $a = 1, b = 7, c = 6, d = 2$

$$p_1 = \frac{d - c}{(a + d) - (b + c)} = \frac{2 - 6}{(1 + 2) - (7 + 6)} = \frac{-4}{3 - 13} = \frac{2}{5}$$

$$p_2 = 1 - p_1 = 1 - \frac{2}{5} = \frac{3}{5}$$

$$q_1 = \frac{d - b}{(a + d) - (b + c)} = \frac{2 - 7}{(1 + 2) - (7 + 6)} = \frac{-5}{3 - 13} = \frac{1}{2}$$

$$q_2 = 1 - q_1 = 1 - \frac{1}{2} = \frac{1}{2}$$

$$V = \frac{a \cdot d - b \cdot c}{(a + d) - (b + c)} = \frac{(1 \times 2) - (7 \times 6)}{(1 + 2) - (7 + 6)} = \frac{2 - 42}{3 - 13} = 4$$

ii. Arithmetic Method

Arithmetic method Steps (Rule)	
Step-1:	Find the difference between the two values of Row-1 and put this value against the Row-2, ignore the sign.
Step-2:	Find the difference between the two values of Row-2 and put this value against the Row-1, ignore the sign.
Step-3:	Find the difference between the two values of Column-1 and put this value against the Column-2, ignore the sign.
Step-4:	Find the difference between the two values of Column-2 and put this value against the Column-1, ignore the sign.
Step-5:	Find probabilities of each by dividing their sum
Step-6:	Find value of the game by algebraic method.

Example-1

1. Find Solution of game theory problem using arithmetic method

Player A \ Player B	B1	B2	B3
A1	10	5	-2
A2	13	12	15
A3	16	14	10

Solution:

1. Saddle point testing

		Player B			
		B1	B2	B3	Row Minimum
Player A	A1	10	5	-2	-2
	A2	13	[12]	15	[12]
	A3	16	(14)	10	10
Column Maximum		16	(14)	15	

Here, Column MiniMax $\neq$ Row MaxiMin . so, this game has no saddle point.

2. Dominance rule to reduce the size of the payoff matrix

		Player B			
		B1	B2	B3	
Player A	A1	10	5	-2	
	A2	13	12	15	
	A3	16	14	10	

Row 1 $\leq$ row 3, so remove row1

		Player B			
		B1	B2	B3	
Player A	A2	13	12	15	
	A3	16	14	10	

Column 1 $\geq$ column 2, so remove column 1

		Player B		
		B2	B3	
Player A	A2	12	15	
	A3	14	10	

Using arithmetic method to get optimal mixed strategies for both the firms.

Player B

Player A

$$\begin{array}{c}
 A_2 \\
 A_3
 \end{array}
 \begin{bmatrix}
 B_2 & B_3 \\
 12 & 15 \\
 14 & 10
 \end{bmatrix}$$

$$\begin{aligned}
 |14 - 10| &= 4 \quad \therefore p_1 = \frac{4}{3+4} = \frac{4}{7} \\
 |12 - 15| &= 3 \quad \therefore p_2 = \frac{3}{3+4} = \frac{3}{7} \\
 |15 - 10| &= 5 \quad \therefore q_1 = \frac{5}{2+5} = \frac{5}{7} \\
 |12 - 14| &= 2 \quad \therefore q_2 = \frac{2}{2+5} = \frac{2}{7}
 \end{aligned}$$

∴ Optimal strategy of A = (4/7, 3/7) ∴ Optimal strategy of B = (5/7, 2/7)

Value of Game

Expected gain of Firm A

(1) $12 \times \frac{4}{7} + 14 \times \frac{3}{7} = \frac{90}{7}$, Firm B adopt B₂

(2) $15 \times \frac{4}{7} + 10 \times \frac{3}{7} = \frac{90}{7}$, Firm B adopt B₃

Expected loss of Firm B

(1) $12 \times \frac{5}{7} + 15 \times \frac{2}{7} = \frac{90}{7}$, Firm A adopt A₂

(2) $14 \times \frac{5}{7} + 10 \times \frac{2}{7} = \frac{90}{7}$, Firm A adopt A₃

Example-2: Find Solution of game theory problem using arithmetic method

Player A \ Player B	B1	B2	B3
A1	1	7	2
A2	6	2	7
A3	5	1	6

Solution:
1. Saddle point testing

		Player B			
		B1	B2	B3	Row Minimum
Player A	A1	1	7	2	1
	A2	(6)	[2]	7	[2]
	A3	5	1	6	1
Column Maximum		(6)	7	7	

Here, Column MiniMax ≠ Row MaxiMin , so the game has no saddle point.

2. Dominance rule to reduce the size of the payoff matrix

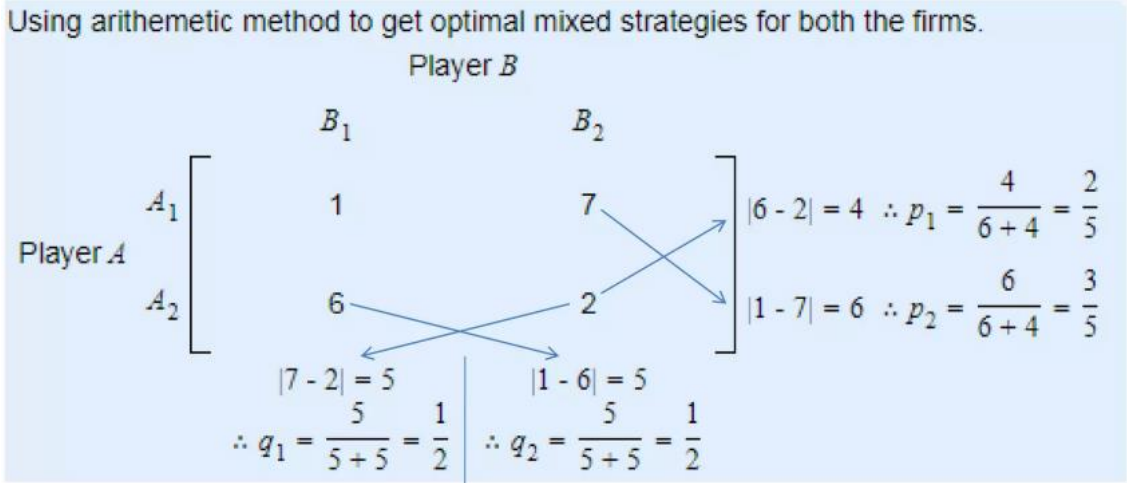
		Player B				
		B1	B2	B3		
Player A	A1	1	7	2		
	A2	6	2	7		
	A3	5	1	6		

Row 3 ≤ row 2, so remove row 3

		Player B				
		B1	B2	B3		
Player A	A1	1	7	2		
	A2	6	2	7		

Column 3 $\geq$ column 1, so remove column 3

		Player B			
		B1	B2		
Player A	A1	1	7		
	A2	6	2		



Hence, Optimal strategy of A = $(\frac{2}{5}, \frac{3}{5})$

Optimal strategy of B = $(\frac{1}{2}, \frac{1}{2})$

Expected gain of Player A

Game value(V) = $1 \times \frac{2}{5} + 6 \times \frac{3}{5} = 4$

Expected Loss of Player B

Game value(V) = $1 \times \frac{1}{2} + 7 \times \frac{1}{2} = 4$

Dominance Theory

Used to simplify a game by eliminating **dominated** strategies (less effective options).

dominance method Steps (Rule)	
Step-1:	If all the elements of Column-i are greater than or equal to the corresponding elements of any other Column-j, then the Column-i is dominated by the Column-j and it is removed from the matrix. eg. If Column-2 $\geq$ Column-4, then remove Column-2
Step-2:	If all the elements of a Row-i are less than or equal to the corresponding elements of any other Row-j, then the Row-i is dominated by the Row-j and it is removed from the matrix. eg. If Row-3 $\leq$ Row-4, then remove Row-3
Step-3:	Again, repeat Step-1 & Step-2, if any Row or Column is dominated, otherwise stop the procedure.
Average dominance method Rule	
1.	If Column-i is dominated by the average of Column-j and Column-k then Column-i is removed from the matrix. eg. If Column-1 $\geq$ average of Column-2 and Column-3, then Column-1 is removed
2.	If Row-i is dominated by the average of Row-j and Row-k then Row-i is removed from the matrix. eg. If Row-1 $\leq$ average of Row-2 and Row-3, then Row-1 is removed

Example-1: Find Solution of game theory problem using dominance method

Player A \ Player B	B1	B2	B3
A1	1	7	2
A2	6	2	7
A3	5	1	6

Solution:
Using dominance property

		Player B				
		B1	B2	B3		
Player A	A1	1	7	2		
	A2	6	2	7		
	A3	5	1	6		

row-3 $\leq$ row-2, so remove row-3

		Player B				
		B1	B2	B3		
Player A	A1	1	7	2		
	A2	6	2	7		

column-3 $\geq$ column-1, so remove column-3

		Player B			
		B1	B2		
Player A	A1	1	7		
	A2	6	2		

Example 2: Find Solution of game theory problem using dominance method

Player A \ Player B	B1	B2	B3	B4
A1	3	5	4	2
A2	5	6	2	4
A3	2	1	4	0
A4	3	3	5	2

Solution:
Using dominance property

		Player B					
		B1	B2	B3	B4		
Player A	A1	3	5	4	2		
	A2	5	6	2	4		
	A3	2	1	4	0		
	A4	3	3	5	2		

Row-3 $\leq$ Row-4, so remove Row-3

		Player B					
		B1	B2	B3	B4		
Player A	A1	3	5	4	2		
	A2	5	6	2	4		
	A4	3	3	5	2		

Column-2 $\geq$ Column-4, so remove Column-2

		Player B				
		B1	B3	B4		
Player A	A1	3	4	2		
	A2	5	2	4		
	A4	3	5	2		

Column-1 $\geq$ Column-3, so remove Column-1

		Player B			
		B3	B4		
Player A	A1	4	2		
	A2	2	4		
	A4	5	2		

Row-1 $\leq$ Row-3, so remove Row-1

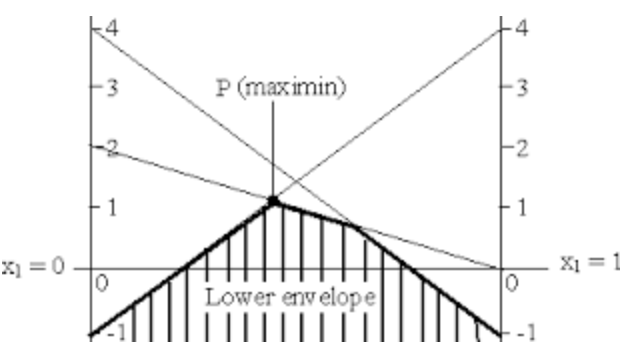
		Player B			
		B3	B4		
Player A	A2	2	4		
	A4	5	2		

Graphical Method

Graphical method Steps (Rule)	
Step-1:	This method can only be used in games with no saddle point, and having a pay-off matrix of type $n \times 2$ or $2 \times n$.
Step-2:	The example is used to explain the procedure.

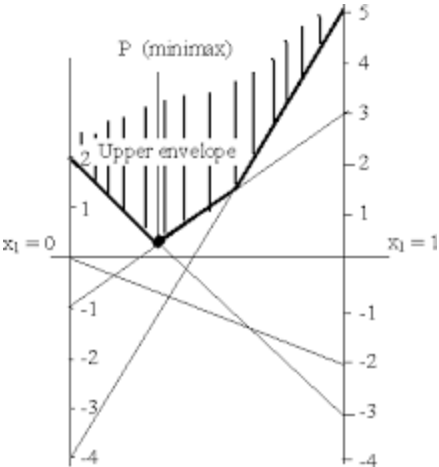
For $2 \times n$

	B1	B2	B3
A1	1	3	12
A2	8	6	2



For $n \times 2$

	B1	B2
A1	-2	0
A2	3	-1
A3	-3	2
A4	5	-4



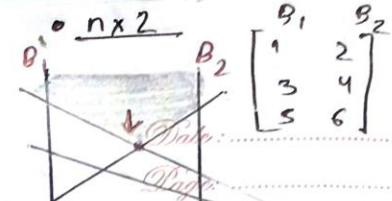
* Graphical Method

→ Useful for $2 \times m$ & $n \times 2$ order

• $2 \times m$

$$A_1 \begin{bmatrix} 1 & 2 & 5 \end{bmatrix}$$

$$A_2 \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$$



→ lowest point at upper envelope

Q.1, Solve the following 2×3 game by Graphical method

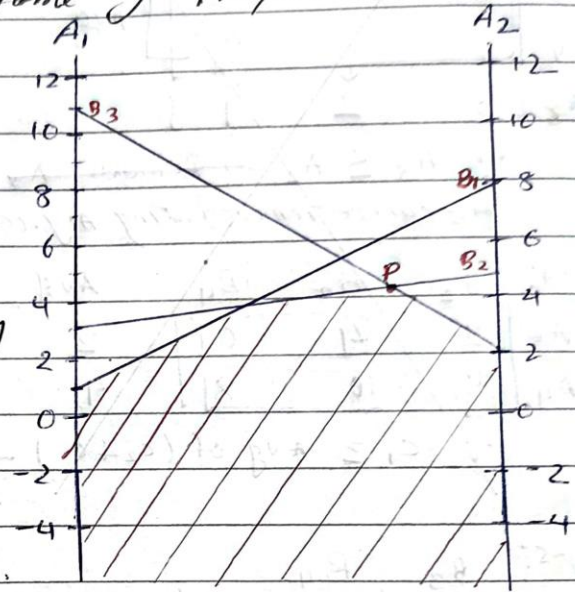
$$A_1 \begin{bmatrix} 1 & 3 & 11 \end{bmatrix}$$

$$A_2 \begin{bmatrix} 8 & 5 & 2 \end{bmatrix}$$

From the graph

The highest point (p) of the shaded region (lower envelope) is the intersecting point of B_2 & B_3 .

So, we consider only two strategies B_2 & B_3 for player B.



Thus, the solution to the original (2×3) game reduces to:

$$A_1 \begin{bmatrix} 3 & 11 \end{bmatrix}$$

$$A_2 \begin{bmatrix} 5 & 2 \end{bmatrix}$$

$$|11-2|=9 \quad |3-5|=2$$

$$\therefore q_1 = 9/11 \quad \therefore q_2 = 2/11$$

$\therefore$ The optimal strategy of A = $\left(\frac{3}{11}, \frac{8}{11} \right)$

" " " B = $\left(\frac{9}{11}, \frac{2}{11} \right)$

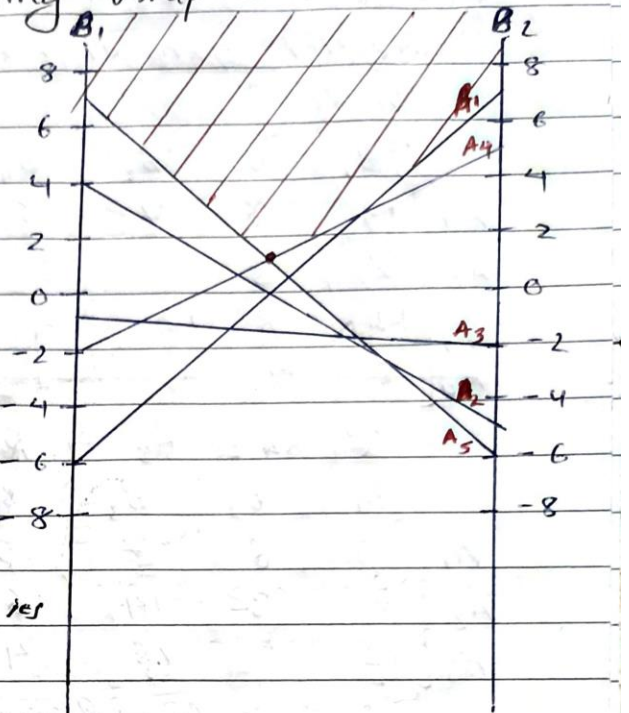
$\therefore$ Value of game (V) = $3 \times \frac{3}{11} + 5 \times \frac{8}{11}$

$$= \frac{9}{11} + \frac{40}{11}$$

$$= \frac{49}{11} \quad //$$

Q.2. Solve the following game using Graphical method.

	B_1	B_2
A_1	-6	7
A_2	4	-5
A_3	-1	-2
A_4	-2	5
A_5	7	-6



From the graph

The lowest point of the shaded region (upper envelope) is the intersecting point of A_4 & A_5

So, we consider only two strategies A_4 & A_5 for player A.

Thus.

$$A_4 \begin{bmatrix} -2 & 5 \end{bmatrix} \begin{matrix} |7-(-6)| = 13 \\ |-2-5| = 7 \end{matrix} \therefore P_1 = \frac{13}{20}$$

$$A_5 \begin{bmatrix} 7 & -6 \end{bmatrix} \begin{matrix} |7-(-6)| = 13 \\ |-2-5| = 7 \end{matrix} \therefore P_2 = \frac{7}{20}$$

$$\therefore q_1 = \frac{11}{20} \quad \therefore q_2 = \frac{9}{20}$$

$$\therefore \text{Optimal strategy of A} = \left(0, 0, 0, \frac{13}{20}, \frac{7}{20} \right)$$

$$\text{Optimal " B} = \left(\frac{11}{20}, \frac{9}{20} \right)$$

$$\therefore \text{Value of game (V)} = -2 \times \frac{13}{20} + 7 \times \frac{7}{20}$$

$$= \frac{-26}{20} + \frac{49}{20}$$

$$= \frac{23}{20} //$$

Q.3. Use the dominance rule to reduce the following game either $2 \times n$ or $n \times 2$ game & then solve by graphical method

	B_1	B_2	B_3	B_4
A_1	19	6	7	5
A_2	7	3	14	6
A_3	12	8	18	4
A_4	8	7	13	-1

$\therefore R_4 \leq R_3 \rightarrow$ Remove R_4

	B_1	B_2	B_3	B_4
A_1	19	6	7	5
A_2	7	3	14	6
A_3	12	8	18	4

$\therefore C_1 \geq C_2 \rightarrow$ Remove C_1 & $C_3 \geq C_4 \rightarrow$ Remove C_3

The reduced matrix is

	B_2	B_4
A_1	6	5
A_2	3	6
A_3	8	4

Now, using graphical method to solve it
From graph

The lowest point from the shaded region is the intersecting point of A_1 & A_2 .

So, we consider only two strategies A_1 & A_2 for Player A.

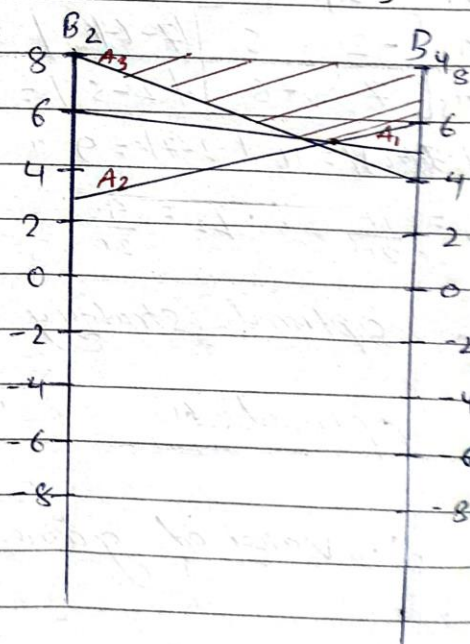
Then,

	B_2	B_4
A_1	6	5
A_2	3	6
	1	3

$\therefore q_1 = 1/4 \quad q_2 = 3/4$

$\therefore$ optimal strategy of A = $(3/4, 1/4, 0, 0)$
B = $(0, 1/4, 0, 3/4)$

$\therefore$ Value of game (V) = $6 \times \frac{3}{4} + 5 \times \frac{1}{4} = 2 \frac{1}{4}$



2023

Past Questions

Q. 12, a. Two types of food manufactures are competing for an increased market share. The payoff matrix describes the increase in market share of A and decrease share of B. Determine optimal strategies for both the manufacturers and the value of the game.

A	B			
	Give prizes	Decrease price	Maintain present strategy	Increase advertising
Give prizes	25	20	14	30
Decrease price	27	16	12	14
Maintain present strategy	35	8	15	19
Increase advertising	-2	8	13	0

Soln

Let the given payoff matrix be:

Step-1	B ₁	B ₂	B ₃	B ₄	Step-2	B ₁	B ₂	B ₃	B ₄
A ₁	25	20	14	30	A ₁	25	20	14	30
A ₂	27	16	12	14	A ₂	27	16	12	14
A ₃	35	8	15	19	A ₃	35	8	15	19
A ₄	-2	8	13	0					

$\therefore C_1 \geq C_2 \rightarrow \text{Remove } C_1$
 $C_4 \geq C_3 \rightarrow \text{Remove } C_4$
 $\therefore R_4 \leq R_3 \rightarrow \text{Remove } R_4$

Step-3	B ₂	B ₃
A ₁	20	14
A ₂	16	12
A ₃	8	15

$\therefore R_2 \leq R_1 \rightarrow \text{Remove } R_1$

Step-4	B ₂	B ₃
A ₂	16	12
A ₃	8	15

$|8-15| = 7 \therefore P_1 = 7/11$
 $|16-12| = 4 \therefore P_2 = 4/11$
 $\therefore \text{optimal strategy of A} = \left(0, \frac{7}{11}, \frac{4}{11}, 0\right)$
 $\therefore Q_1 = 3/11 \therefore Q_2 = 8/11$
 $\therefore \text{optimal strategy of B} = \left(0, \frac{3}{11}, \frac{8}{11}, 0\right)$

$$\therefore \text{The value of game (V)} = 16 \times \frac{7}{11} + 8 \times \frac{4}{11} = \frac{144}{11}$$

2024

Date:

Pages:

Q. 4 Reduce the following two person zero sum game to 2×2 order by dominance rule and obtain the optimal strategies for each player and the value of the game

Player A	Player B			
	B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

Soln:

Since, $R_1 \leq R_3 \rightarrow$ Remove R_1 ,

	B_1	B_2	B_3	B_4
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

Since, $C_1 \geq C_3 \rightarrow$ Remove C_1 ,

	B_2	B_3	B_4	Avg		B_3	B_4	
A_2	4	2	4	3	$\Rightarrow$	A_2	2	4
A_3	2	4	0	2		A_3	4	0
A_4	4	0	8	4		A_4	0	8
$\therefore C_1 \geq \text{Avg of } (C_2, C_3) \rightarrow \text{Remove } C_1$						Avg	2	4

$\therefore C_1 \geq \text{Avg of } (C_2, C_3) \rightarrow$ Remove C_1

Now, using Arithmetic method to solve the problem

	B_3	B_4	
A_3	4	0	8
A_4	0	8	4
	8	4	

$$\therefore P_1 = \frac{8}{12} = \frac{2}{3}$$

$$\therefore P_2 = \frac{4}{12} = \frac{1}{3}$$

$$Q_1 = \frac{2}{3} \quad Q_2 = \frac{1}{3}$$

$\therefore$ Optimal strategy of A = $\left(\frac{2}{3}, \frac{1}{3}\right)$

"

$$B = \left(0, 0, \frac{2}{3}, \frac{1}{3}\right)$$

$$\therefore \text{value of game (V)} = 4 \times \frac{2}{3} + 0 = \frac{8}{3}$$